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# Introduction to orbits and black holes

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# 1 Introduction

For the teachers, these notes aim at illustrating how to make the most out of a classic undergraduate problem. From the calculation of orbits, many examples and applications of the mathematical tools from the high school syllabus can be applied, and many concepts interconnected.

For the students, mI want them to learn how to tackle a real theoretical problem using all the mathematical tools at their disposal. There is immense power and satisfaction in being able to prove basic principles to ourselves, such as the conservation of angular momentum or energy. Physics and Mathematics are tools, meant to be at your service, and not the other way around.

The main source of inspiration for the course is chapter 4 of [1]. The section on Einstein's corrections comes from a mix of [2] and [3]. From these references, I take the structure and how certain concepts are presented, adapting them to my style. For the black holes part, I rely on examples and theory from [3], a simplified version of [4]. A very well-known set of notes in Spanish available to everyone are those by Bert Janssen. For more detailed books and notes, I recommend consulting the introductory sections of the references.

To accompany these notes, I have designed three worksheets with exercises, to give examples of how to translate theoretical knowledge into activities with the students. Worksheets 1 and 2 aim at introducing Python as a tool for plotting functions. In this context, Python is not presented as an additional cognitive load, but as a virtual laboratory that allows students to experiment with Physics. By integrating programming in the classroom, you transform static formulas into dynamic simulations, allowing students to 'touch' the variables of an orbit and observe their consequences in real-time. This transition from paper to screen not only increases motivation but equips the students with an essential technological skill, turning learning into a process of empirical and visual discovery. Worksheet 3 focuses on simple analytical calculations. Although General Relativity is complex, there are numerous results that can be used to contextualize the calculation of a derivative or isolating a variable.

Lastly, I have left some incomplete calculations as exercises. This serves two purposes: to motivate you to get acquainted with the notation and to provide additional examples of simple calculation exercises that they can work on with the students. They are never more than a few lines long, so I highly recommend giving them a go.

## 1.1 Notation

Throughout these notes, I use dots ($\bullet$) to represent derivatives with respect to time. Multiple dots are equivalent to multiple derivatives, that is

$$\dot{x} = \frac{dx}{dt}, \quad \ddot{x} = \frac{d^2x}{dt^2}. \tag{1.1}$$

To represent vectors, I underline the corresponding variable or function. For example, the 3-dimensional Cartesian position vector is written as $\underline{r} = (x, y, z)$, where $x, y, z \in \mathbb{R}$. Finally, I denote unit vectors with a circumflex or hat over the vectors, such that $\underline{\hat{x}}$ is the vector pointing in the direction of the $x$-axis, with a magnitude of $|\underline{\hat{x}}| = 1$.

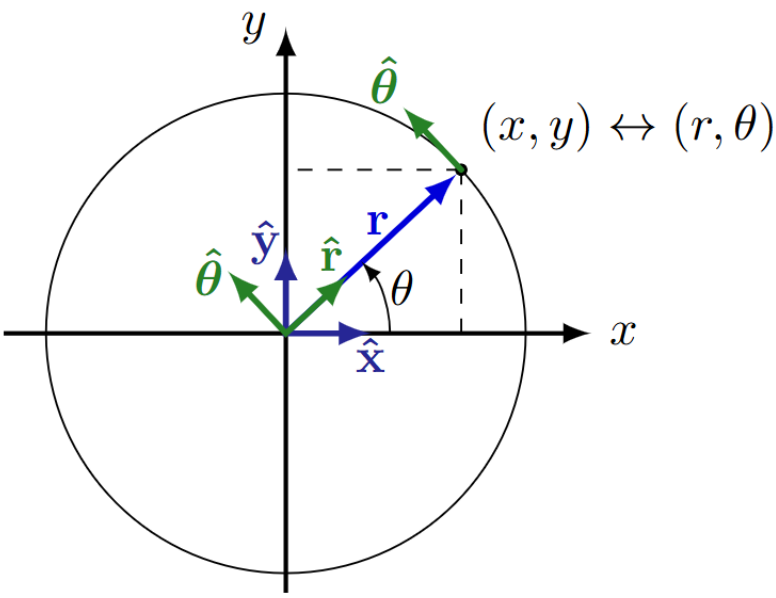


**Fig. 1**: Representation on the plane of Cartesian coordinates $(x, y)$, the Cartesian unit vectors $(\underline{\hat{x}}, \underline{\hat{y}})$, polar coordinates $(r, \theta)$, and their respective unit vectors $(\underline{\hat{r}}, \underline{\hat{\theta}})$. Based on code by Izaak Neutelings.

### 1.2 Preamble: polar coordinates

Orbital motions, as we will demonstrate next, always occur on a plane that does not change its orientation. This reduces our problem from 3 to 2 dimensions, because we can align the $z$-axis of our coordinates with the normal to the plane, and deal only with $(x, y)$.

The best way to exploit the symmetry of the problem is to transform the coordinates $(x, y)$ to describe the plane with *polar coordinates* $(r, \theta)$, given by

$$x = r\cos\theta, \quad y = r\sin\theta. \tag{1.2}$$

The coordinate $r \in (0, \infty)$ gives the radial distance to the origin, and $\theta \in [0, 2\pi)$ gives the angle of the radial distance with the $x$-axis, measured counterclockwise. Fig. 1 shows a graphical representation of these coordinates.

With these basic definitions in mind, we can define two unit vectors that allow us to represent any vector in the plane.

$$\underline{\hat{r}} = (\cos\theta, \sin\theta), \quad \underline{\hat{\theta}} = (-\sin\theta, \cos\theta). \tag{1.3}$$

**Exercise 1.1.** *Prove that*

1. *Any point $(a, b)$ on the $(x, y)$ plane can be expressed as*

$$(a, b) = c_1\underline{\hat{r}} + c_2\underline{\hat{\theta}}, \tag{1.4}$$

   *where $c_1$ and $c_2$ are two real constants.*

2. *Both $\underline{\hat{r}}$ and $\underline{\hat{\theta}}$ are unit vectors for all $\theta$, meaning $|\underline{\hat{r}}| = |\underline{\hat{\theta}}| = 1$.*

*3. They are orthogonal vectors for all $\theta$, meaning*

$$\hat{\underline{r}} \cdot \hat{\underline{\theta}} = 0. \tag{1.5}$$

Having these three properties, we know we can use $(\hat{\underline{r}}, \hat{\underline{\theta}})$ as unit vectors to describe the plane, as an alternative to $(\hat{\underline{x}}, \hat{\underline{y}})$. The mathematical term is that they constitute a *basis*.

However, some properties are very different. The Cartesian unit vectors are constant, even if the body of interest moves: $\dot{\hat{\underline{x}}} = 0$ and $\dot{\hat{\underline{y}}} = 0$. But looking at the definitions (1.3), we see that this is not the case for the polar unit vectors. In fact, if our starting point for the position of a body is

$$\underline{r} = r\hat{\underline{r}}, \tag{1.6}$$

we can calculate the velocity in polar coordinates using the chain rule

$$\begin{aligned} \dot{\underline{r}} &= \frac{d}{dt}(r\hat{\underline{r}}) = \dot{r}\hat{\underline{r}} + r\frac{d\hat{\underline{r}}}{dt} \\ &= \dot{r}\hat{\underline{r}} + r\frac{d\hat{\underline{r}}}{d\theta}\frac{d\theta}{dt} \\ &= \dot{r}\hat{\underline{r}} + r\dot{\theta}\hat{\underline{\theta}}. \end{aligned} \tag{1.7}$$

The second term appears precisely because of the change of our basis $(\hat{\underline{r}}, \hat{\underline{\theta}})$ over time, and is proportional to the *angular velocity* $\dot{\theta} = \omega$. This angular velocity is the same one we defined when studying uniform circular motion (UCM). We can take the derivative with respect to time again to find the acceleration.

**Exercise 1.2.** *Starting from* (1.7)*, show that*

$$\ddot{\underline{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{\underline{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\underline{\theta}}. \tag{1.8}$$

*Once again, we find quantities that appear in the definitions of circular motion, in this case uniformly accelerated circular motion (UACM). If the radius is constant ($\dot{r} = \ddot{r} = 0$) and the angular acceleration is constant ($\ddot{\theta} = 0$), the magnitude of the acceleration is $|\ddot{\underline{r}}| = r\dot{\theta}^2 = r\omega^2 = v^2/r$, where $v$ is the tangential velocity $v = r\omega$. This is the centripetal acceleration.*

After this preamble, we are ready to tackle the problem of orbits.

## 2 Newtonian Mechanics and the study of orbits

### 2.1 The Law of Universal Gravitation

The fascination with the sky and the bodies that inhabit it has been part of humanity's history since its beginnings. We started by identifying celestial bodies with spirits and gods. Then, we carefully studied these, as a means to organize our communities and find our place on the universe and on planet Earth. One of the first and most extensive catalogs of stars and constellations dating back to around 1000 B.C., the *Mul.Apin* series [5]. In the

3rd century B.C., Eratosthenes used the shadows cast by the Sun to estimate the radius of the Earth [6]. But the most important shift occurs in the 17th century with the arrival of Newton and his laws of motion and universal gravitation. Since then, we transitioned from observing and estimating to predicting, with an accuracy that remains impressive to this day.

Newton´s achievements cannot be overstated. Standing *on the shoulders of giants*, his work laid the foundations and mathematical tools for classical mechanics, which until the 19th century explained all physical phenomena with its laws. Specifically, the movement of the celestial bodies inhabiting our sky is explained by a single unifying principle: *gravity*. Through Newton's mathematical description, we can understand the shape of planetary orbits around the Sun, the tides, and even estimate how much our planet weighs. Newton's work is precisely the starting point of this module: his Second Law and the Law of Universal Gravitation. To simplify our work, we will assume that all planets and the Sun are point masses. Our initial goal is to describe the motion of a body of mass $m$, due to gravity, around a fixed body of mass $M$ which we place at the origin. A concrete example would be the Earth orbiting around the Sun. To describe its trajectory due to gravity, we will use 3 spatial dimensions $(x, y, z)$ and one temporal dimension $t$. Taking all this into account, the equation where everything starts is

$$m\ddot{\underline{r}} = -\underline{\nabla} V(r), \tag{2.1}$$

where $V$ is the gravitational potential generated by the motionless body, that is, our Sun. This potential only depends on the distance to the origin $r = |\underline{r}| = \sqrt{x^2 + y^2 + z^2}$, with $\underline{r} = (x, y, z)$ being the position vector of a body from the origin. As expected, $r \in (0, \infty)$.

The symbol "$\nabla$" appearing in (2.1) is known as the gradient. It maps scalar function into vectors using derivatives. We are independently applying Newton's second law on each of our 3 Cartesian axes, so that

$$m\ddot{\underline{r}} = -\underline{\nabla} V(r) \iff m\left(\ddot{x}, \ddot{y}, \ddot{z}\right) = -\left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\right) \iff \begin{cases} m\ddot{x} = -\frac{\partial V}{\partial x}, \\ m\ddot{y} = -\frac{\partial V}{\partial y}, \\ m\ddot{z} = -\frac{\partial V}{\partial z}. \end{cases} \tag{2.2}$$

Here, the symbols $\partial/\partial_x$, $\partial/\partial_y$, and $\partial/\partial_z$ denote partial derivatives, which for our purposes work the same as normal ones, treating the rest of the variables as constants. The gradient can also be expressed in a much more compact and convenient way.

**Exercise 2.1.** *Show that*

$$\underline{\nabla} V(r) = \frac{dV}{dr}\hat{\underline{r}}, \tag{2.3}$$

*where* $\hat{\underline{r}} = \underline{r}/r$.

Our first step in calculating orbits will be to simplify the problem as much as we can. This is how Physics often works, going from the general to the specific. Recalling the definition of angular momentum

$$\underline{L} = m\underline{r} \times \dot{\underline{r}}, \tag{2.4}$$

we can prove that it is conserved. If we take the derivative with respect to time, using the product rule (which also works for a cross product), we have

$$\dot{\underline{L}} = m\dot{\underline{r}} \times \dot{\underline{r}} + m\underline{r} \times \ddot{\underline{r}}. \tag{2.5}$$

The first term is zero because the cross product of two parallel vectors is 0. For the second term, using (2.3), we get

$$\dot{\underline{L}} = m\dot{\underline{r}} \times \dot{\underline{r}} + m\underline{r} \times \ddot{\underline{r}} = 0 - \frac{dV}{dr}\underline{r} \times \hat{\underline{r}} = 0, \tag{2.6}$$

as we wanted to show.[1] Furthermore, by the definition of $\underline{L}$ we know that

$$\underline{r} \cdot \underline{L} = m\underline{r} \cdot (\underline{r} \times \dot{\underline{r}}) = 0, \tag{2.7}$$

since $\underline{r} \times \dot{\underline{r}}$ is perpendicular to $\underline{r}$. Therefore, the position of the body of mass $m$ is always on a plane perpendicular to $\underline{L}$. Using the same argument, $\dot{\underline{r}} \cdot \underline{L} = 0$, so the velocity of the body of mass $m$ is also on the plane. What if we try to rewrite the problem so that everything occurs on a single plane instead of in 3 dimensions?

First, we align our $z$ with the angular momentum $\underline{L}$. Then, using (2.3) and polar coordinates $(r, \theta)$ (see section 1.2 for an introduction), we can express (2.1) as

$$m(\ddot{r} - r\dot{\theta}^2)\hat{\underline{r}} + m(r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\underline{\theta}} = -\frac{dV}{dr}\hat{\underline{r}}. \tag{2.8}$$

Then, we have two separate equations, one for each polar direction

$$\ddot{r} - r\dot{\theta}^2 = -\frac{dV}{dr}, \tag{2.9}$$

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0. \tag{2.10}$$

Using the product rule in reverse on (2.10), we have

$$\begin{aligned} & r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 \\ \iff & \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right) = 0, \\ \iff & r^2\dot{\theta} = l, \end{aligned} \tag{2.11}$$

where $l$ is a name given to a constant. In fact, we already know this constant.

**Exercise 2.2.** *Show, using* (2.4) *and polar coordinates, that*

$$L = |\underline{L}| = ml. \tag{2.12}$$

This implies that $l$ is the angular momentum per unit mass, and equation (2.10) is none other than the conservation of angular momentum. Going back to (2.9) and using $l = r^2\dot{\theta}$, we obtain

$$m\ddot{r} = -\frac{dV}{dr} + \frac{ml^2}{r^3}. \tag{2.13}$$

[1] From here you can also extract the definition of torque $\underline{\tau}$. If instead of (2.1) we reach $m\ddot{\underline{r}} = \underline{F}$, where $\underline{F}$ is an arbitrary force, we have $\dot{L} = \underline{r} \times \underline{F} = \underline{\tau}$.

This is the only equation we have to solve. A more compact and useful way to express (2.9) is to use the chain rule in reverse

$$m\ddot{r} = -\frac{dV}{dr} + \frac{ml^2}{r^3} = -\frac{d}{dr}\left(V(r) + \frac{ml^2}{2r^2}\right) = -\frac{d}{dr}V_{\text{eff}}(r), \tag{2.14}$$

where

$$V_{\text{eff}}(r) = V(r) + \frac{ml^2}{2r^2}. \tag{2.15}$$

$V_{\text{eff}}(r)$ is known as the effective potential, with $ml^2/2r^2$ being the *angular momentum barrier* or *centrifugal barrier*. It prevents the body from getting too close to the origin, as we will see next.

It is worth stopping here to reflect on what we have done qualitatively. Starting from the assumption that the potential is radial, we've used the system's symmetries to go from 3 dimensions (something super complex) to a single equation of one variable. What we just did is a ***FUNDAMENTAL*** process in theoretical physics: symmetries give rise to constants or conservation laws, and conservation laws allow us to simplify and solve problems.

When there are not enough symmetries or conserved quantities, the problems get complicated. For this reason, problems like the $N$-body problem ($N$ bodies gravitationally interacting with each other) for $N \geq 3$ have no analytical solution.

## 2.2 Analytical approach to the orbit

Our goal is to solve

$$m\ddot{r} = -\frac{dV}{dr} + \frac{ml^2}{r^3} \tag{2.16}$$

given the potential $V(r)$. This can also be the starting point for the orbit problem if we want to skip the geometric explanation (or save it for later). Before substituting Newton's $V(r)$, we can learn from the problem (and apply many calculus tools) by looking at the system's energy $E$. By the definition of energy and (1.7), we have

$$\begin{aligned} E &= \frac{1}{2}m\underline{\dot{r}} \cdot \underline{\dot{r}} + V(r), \\ &= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + V(r), \\ &= \frac{1}{2}m\dot{r}^2 + \frac{ml^2}{2r^2} + V(r), \\ &= \frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r). \end{aligned} \tag{2.17}$$

The system's energy can be expressed in terms of kinetic energy added to an effective potential, which is why we defined it in (2.15). And since nothing depends on time, the energy is conserved.

**Exercise 2.3.** *Using (2.14), show that the energy $E$ defined in (2.17) is conserved:*

$$\dot{E} = \frac{dE}{dt} = 0. \tag{2.18}$$

Knowing that $E$ is a constant, we could isolate $\dot{r}^2$ in (2.17) and try to solve (for $m \neq 0$)

$$\dot{r} = \frac{dr}{dt} = \pm\sqrt{\frac{2}{m}}\sqrt{E - V_{\text{eff}}(r)}. \tag{2.19}$$

Then, using the fundamental theorem of calculus (integration), we have

$$t = \pm\sqrt{\frac{m}{2}}\int \frac{dr}{\sqrt{E - V_{\text{eff}}(r)}}. \tag{2.20}$$

The problem is that, except for some specific $V_{\text{eff}}(r)$, this integral is very complex and may not even have an analytical solution [1].[2]

Something much more feasible is to study $V_{\text{eff}}(r)$. We will assume we have Newtonian gravity, meaning the potential is given by the Law of Universal Gravitation

$$V(r) = -\frac{mk}{r}, \tag{2.21}$$

for $k = GM > 0$, with $G \sim 6.67 \times 10^{-11}\text{N} \cdot \text{m}^2/\text{kg}^2$ and $M$ being the mass of the attracting body. Using the potential (2.21), we have

$$V_{\text{eff}}(r) = -\frac{mk}{r} + \frac{ml^2}{2r^2}. \tag{2.22}$$

Knowing that energy is conserved, plotting $V_{\text{eff}}(r)$ allows us to learn about the problem before solving it. The minimum of $V_{\text{eff}}(r)$ is at $r_{\text{min}} = l^2/k$ and gives us $V_{\text{eff}}(r_{\text{min}}) = -mk^2/2l^2$. Depending on $E$, we have different types of orbits.

1. If $E = E_{\text{min}} = -mk^2/2l^2$, the body stays at $r = r_{\text{min}}$. But, recall that $\dot{\theta} = l/r^2 = l/r_{\text{min}}^2$, so

$$\theta = \frac{l}{r_{\text{min}}^2}t + \theta_0, \tag{2.23}$$

   where $\theta_0$ is a constant. We thus have a circular orbit.

2. If $E_{\text{min}} < E < 0$, the body stays in the potential valley $V_{\text{eff}}(r)$, but both $r$ and $\theta$ vary with time. It is not at all obvious, but we'll see next that for $V(r) = -mk/r$ this describes an *elliptical* orbit.

3. If $E > 0$, the body can be above the horizontal axis, as observed in Fig. 2. It approaches from infinity, reaches a minimum distance $r$, and then goes back off into infinity. For $V(r) = -mk/r$ this describes a *hyperbolic* orbit.

Plotting (2.22) is an interesting calculus exercise that to apply all typical continuity and limit tools. This is what the first part of practical session 1 consists of. Having studied the energies and gaining intuition for what we expect to find, we now attack the problem.

[2] It can be solved using numerical methods like Euler's method or the more advanced Runge-Kutta. We'll leave that for another module :)

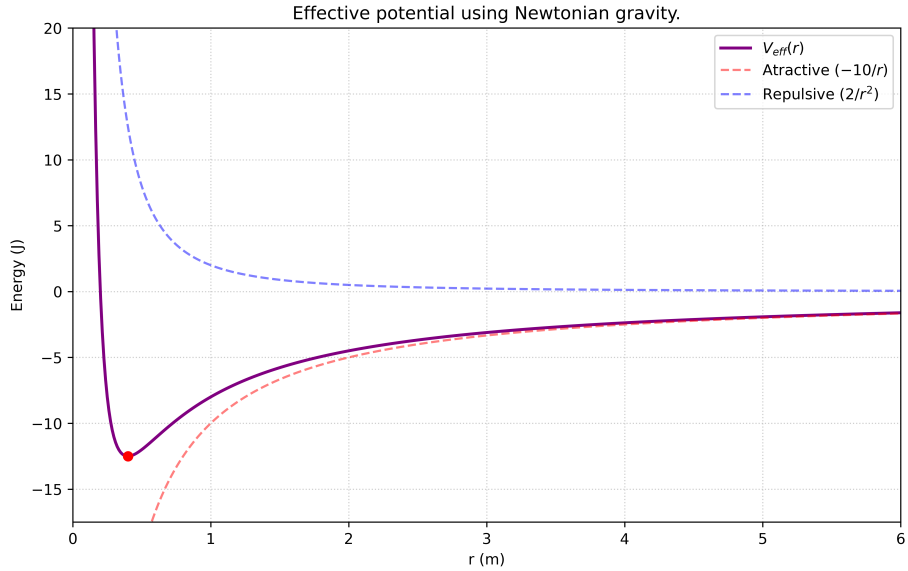


**Fig. 2**: Representation of the potential (2.22), for $k = 10$, $m = 1$, and $l = 2$ (solid violet line). Also depicted are the attractive term (dashed red line) and the repulsive term (dashed blue line). The minimum of the function $V_{\text{eff}}(r)$ is indicated by a red circle.

Solving for $r(t)$ and $\theta(t)$ analytically is complicated: you have to solve (2.20). Instead of $r(t)$ and $\theta(t)$, we will try to obtain a solution that tells us about the orbit geometrically, that is, $r(\theta)$. The first step here is a mathematical trick. There is no geometric motivation according to Tong [1], but it works. Letting

$$u = \frac{1}{r}, \tag{2.24}$$

the problem is greatly simplified. To see this, apply the chain rule to rewrite (2.13).

**Exercise 2.4.** *Demonstrate, using the chain rule and $u = 1/r$, that*

$$\dot{r} = \frac{dr}{dt} = -l\frac{du}{d\theta}, \tag{2.25}$$

$$\ddot{r} = \frac{d^2r}{dt^2} = -l^2u^2\frac{d^2u}{d\theta^2}. \tag{2.26}$$

Taking these changes into account, equation (2.13) becomes

$$\frac{d^2u}{d\theta^2} + u = -\frac{1}{ml^2}\frac{dV}{du}, \tag{2.27}$$

This equation is known as the *orbit equation.*[3] Substituting $V(u) = -mku \implies dV(u)/du = -mk$ into (2.13), we get

$$\frac{d^2u}{d\theta^2} + u = \frac{k}{l^2}. \tag{2.28}$$

[3]Once we have $u(\theta)$, everything's time dependence can be calculated using $\dot{\theta} = lu^2$, using integration.

But we have already seen this problem before. In fact, it is almost the only problem we solve in classical mechanics: the *harmonic oscillator*. What happens is that its center is shifted. Explicitly, defining $\tilde{u} = u - k/l^2$, (2.28) becomes

$$\frac{d^2\tilde{u}}{d\theta^2} + \tilde{u} = 0. \tag{2.29}$$

Then, the solution to our problem is

$$u(\theta) = A\cos(\theta - \theta_0) + \frac{k}{l^2}, \tag{2.30}$$

with $A$ and $\theta_0$ being constants.[4] Returning to polar coordinates, we get

$$r(\theta) = \frac{r_0}{1 + e\cos\theta}. \tag{2.31}$$

Here we have defined the constants $r_0 = l^2/k$ and the *eccentricity* $e = Al^2/k$. This describes conic sections, and another suggested exercise is to draw these using Python, as we do in worksheet 1.

**Exercise 2.5.** *Demonstrate that different values of $e$ yield different conic sections, by writing (2.31) in Cartesian coordinates. Specifically*

1. *$e = 0$ describes a circle.*
2. *$0 < e < 1$ describes an ellipse.*
3. *$e = 1$ describes a parabola.*
4. *$e > 1$ describes a hyperbola.*

Finally, we can go back to our energy equation and substitute everything we have learned. Using the substitution $u = 1/r$ in (2.17) and the chain rule, we have

$$E = \frac{mk^2}{2l^2}(e^2 - 1). \tag{2.32}$$

And comparing with our previous results, we find that

1. $e < 1 \implies E < 0$, these are bound orbits, ellipses.
2. $e > 1 \implies E > 0$, these are unbound orbits, which we now know are hyperbolas.
3. $e = 0 \implies E = -mk^2/2l^2$, which are circular orbits.
4. $e = 1 \implies E = 0$, marking the transition between bound and unbound, which are parabolas.

[4] This is completely equivalent to $u = A\cos\theta + B\sin\theta + \frac{k}{l^2}$, but this form is convenient for us to simplify isolating $u(\theta) = 1/r(\theta)$

We have now solved the problem we set out to attack! Before making life a bit harder for ourselves, we will take a break to catch our breath and demonstrate the power of Newtonian gravity.

**Exercise 2.6.** *Kepler's laws date from 1605, and can be deduced using Newton's laws. Using the results of our calculations, convince yourself that they are true.*

1. *The planets of the solar system orbit in ellipses, with the Sun at one of their foci.*
2. *The line joining any planet to the Sun sweeps out equal areas in equal intervals of time.*
3. *The square of the orbital period is proportional to the cube of the radius (or period proportional to the radius to the power of 3/2).*

*A hint for the last one: dimensional analysis.*

Now we shift gears and give gravity a new twist. We delve into the world of General Relativity.

## 3 Einstein's gravity and its corrections

For very massive objects, Newton's Law of Universal Gravitation fails to describe reality correctly and we must use the theory of General Relativity, in which Einstein describes gravity as the curvature of spacetime. To use General Relativity in depth, one must apply tools from differential geometry, a beautiful area of mathematics from the final years of an undergraduate degree. However, for certain simple problems or when gravity is not extremely strong, we can use Newton's theory of gravity with some modifications called *corrections*, and arrive at incredibly powerful proofs. This is one of the great triumphs of General Relativity: when gravity is weak, it reproduces Newton's mechanics accurately.

### 3.1 Black holes: light has no escape.

Informally, a black hole is a region of space where gravity is so strong that not even light can escape from it. Although the definition may seem counterintuitive at first, it can be motivated using a theoretical example with Newtonian mechanics.

Let's imagine we work at NASA and want to put a satellite into orbit, launching it into space from the Earth's surface. We will assume the only force slowing the satellite down is the gravitational pull, meaning we ignore air resistance and any other interactions. We will also assume our final goal is for the satellite to wander through space, that it has no angular momentum, and that there are no other celestial bodies. With all these ingredients, we can use the principle of energy conservation. Using (2.22) with $l = 0$, at the Earth's surface $r = R_{\text{earth}}$ and for a satellite with mass $m$ shot out at a velocity $v_0$, we have

$$E_{\text{earth}} = \frac{1}{2}mv_0^2 - \frac{mk}{R_{\text{earth}}}. \tag{3.1}$$

If we finally succeed in having the satellite wander through outer space, the potential term is 0 and we want the satellite to just "stop" (because our goal is just to escape Earth's gravitational pull)

$$E_\infty = \frac{1}{2}m \times 0^2 - 0 = 0. \tag{3.2}$$

But using the conservation of energy, we have $E_{\text{earth}} = E_\infty = 0$. Then, solving for $v_0$, we have

$$E_{\text{earth}} = 0 \implies v_0 = \sqrt{\frac{2k}{R_{\text{earth}}}} = \sqrt{\frac{2GM_{\text{earth}}}{R_{\text{earth}}}}. \tag{3.3}$$

The quantity $v_0$ is known as the *escape velocity*, and it tells us how fast a body must be launched from Earth's surface so that it ends up wandering freely through outer space. The interesting thing is that the escape velocity does not depend on the mass of the object in question, but on the mass of the attracting body and the distance to it.

**Exercise 3.1.** *At what speed should I launch the satellite so it leaves Earth's surface? Now suppose the Earth has the same mass, but its radius is halved. At what speed would the satellite have to be launched in that case?*

This theoretical experiment of shrinking the Earth can be taken further. The more we compact the Earth while keeping the same mass, the higher the escape velocity or the speed at which I have to launch the satellite. When the radius of the Earth is approximately that of a sugar cube, something strange happens. For $r \sim 10\ mm$, we have

$$v_0 = \sqrt{\frac{2GM_{\text{earth}}}{R_{\text{earth}}}} \sim 3 \times 10^8\ m/s. \tag{3.4}$$

This is the speed of light. For such a concentration of mass, not even light would be able to escape from the Earth's surface. Back in the late 18th century, a century after Newton's gravity (1687), an astronomer named John Mitchell devoted himself to calculating the escape velocity of stars of various sizes and masses. In his calculations, he realized that if a star was dense enough, the escape velocity of the star is the speed of light. A star would generate light and expel it into outer space, but the gravitational pull would cause it to turn around halfway and fall back to the star. For this reason, he named this type of star a *dark star*. Thirteen years later, the astronomer Laplace popularized the same prediction in "Exposition of the System of the World" (independently of Mitchell's work). Curiously, in the third edition of his work, in 1808, he omitted the mention of dark stars [7]. Right around that time, experiments were coming to light (pun intended), speaking of light as a wave. People were unsure about how to interpret light's interaction with Newtonian gravity.[5]

[5] Nowadays, we know that light behaves as both, particle and wave, and depending on the context we use one perspective or the other. E.g., Young's double-slit experiment (1802) requires thinking of light as a wave. The photoelectric effect, discovered by Hertz in 1887 and explained by Einstein in 1905, is easier to explain if you think of light as a particle, a photon.

### 3.2 The Schwarzschild solution

About 100 years had to pass for black holes to go from being a counterintuitive idea to a viable theoretical example. This happens with the arrival of General Relativity in 1915. Such theory was a massive revolution, and since then, black holes were the source of many debates. Einstein developed a physical theory describing how the mass of objects changes the space around them and conditions the movement of matter, including light. To understand how light interacts with gravity, one has to solve equations known as Einstein's field equations. The problem is that the equations are extremely complicated. In fact, to this day, we continue to study them because we are far from having all the solutions under control.

The first exact solution was found by Karl Schwarzschild. Given how complicated Einstein's equations are, he looked for a solution by simplifying the problem using symmetries (as we did in section 2). He assumed spherical symmetry (meaning that rotating the attracting body changes nothing around it) and that there were no further additions: no other masses, no charge, no rotation. A few months after Einstein published his equations, he managed to solve them exactly. The Schwarzschild metric describes the space around a spherical object with mass that is completely stationary. It is characterized by the function

$$f(r) = 1 - \frac{2GM}{rc^2}, \tag{3.5}$$

where $M$ is the mass of the attracting body and $c$ is the speed of light.[6] Observing (3.5), we see that $f$ is not defined for $r = 0$. For mathematical reasons, we cannot have $f(r) = 0$ either.[7] However, we have that

$$f(r_s) = 0 \iff r_s = \frac{2GM}{c^2}. \tag{3.6}$$

So it seems that $r_s$ is also a special point to keep in mind. If we want to describe, for example, planetary orbits around the Sun, everything is great, because we don't need to consider $r = r_s$. Substituting the mass of the Sun, we have that

$$r_s = \frac{2GM}{c^2} \sim 3km \ll R_\odot \sim 7 \times 10^5 km. \tag{3.7}$$

So as long as we are beyond the surface of the Sun, that is $r > r_s$, we can use the Schwarzschild metric without any problem, because $f(r)$ neither becomes zero nor diverges.

Nonetheless, something strange happens when we consider very dense spherical objects, with very large and concentrated masses, just as Mitchell and Laplace did. If $r_s$ is greater than the radius of the celestial body in question, we are forced to consider the Schwarzschild solution for $r = r_s$ and take into account what happens at this point. Today, we know that in these cases the Schwarzschild metric describes the simplest theoretical example of a black hole.

[6] In General Relativity, physicists are quite lazy and work with *natural units*. In these units, $c = 1$, so it is quite common to find this function written as $f(r) = 1 - 2GM/r$. There are people who even commit the audacity of also fixing $G = 1$, so that $f(r) = 1 - 2M/r$.

[7] Metrics in General Relativity are rulers for measuring distances in space and time. When we write the Schwarzschild metric using $f(r)$, the notion of "radial" distance is proportional to $1/f(r)$, while the "temporal" distance is proportional to $f(r)$. When $f(r) = 0$, this rule stops making physical sense.

In Fig. 3 we have a pictorial representation of a Schwarzschild black hole. The grayish area indicates the region from which light, if it gets there, does not escape. We call the region delimiting the point of no return the *event horizon*, and it is nothing more and nothing less than the radial distance $r = r_s$. We call the central point of the black hole the *singularity*. Studying $f(r)$ we can intuit that something terrible happens here with our mathematical model, since $|f(r)| \to \infty$ as $r \to 0$.

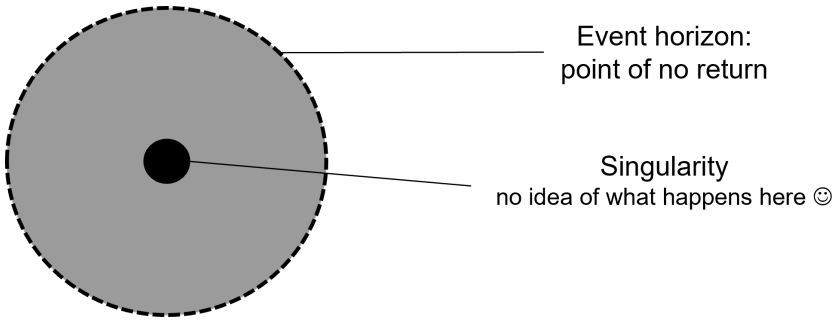


**Fig. 3**: Schematic representation of a Schwarzschild black hole. The event horizon $r_s = 2GM/c^2$ delimits the interior of the black hole in gray. The singularity $r = 0$ is represented by a black circle.

This is the picture we have in mind right now, but back in the day, many physicists were confused about how to interpret the Schwarzschild solution. Until there was theoretical consensus on the existence of black holes in nature and that they could actually form, about 40 years passed.[8] Einstein, along with Eddington, were among the main detractors of the existence of black holes in nature.

The fundamental problem was that many scientists' intuition wanted to protect nature from such strange objects. How could so much mass naturally concentrate in so little space? The answer is that stellar-mass black holes (up to about 100 times the mass of the Sun) form after the death of stars with sufficient mass. A normal star, like our Sun, is a ball of burning matter that exists thanks to the balance of two effects: gravity, which tries to pull all the matter that makes up the star to the center, and the pressure generated by nuclear reactions inside the star. These reactions are what cause the star to burn and emit light, and prevent it from collapsing in on itself. While nuclear reactions occur, the matter that makes up the star also acquires a massive amount of energy and tries to shoot out in all directions. When the star runs out of fuel or cannot burn it efficiently, that pressure from nuclear fusion ceases to exist and it begins to collapse due to gravity.

Einstein, Eddington, and other astronomers thought that as the star collapsed on itself, some physical process would have to stop or prevent such collapse. The idea of a region of space completely disconnected from the outside world and from which light could not escape was counterintuitive. Nonetheless, it was also truthful. Today, thanks to the work of many scientists like Oppenheimer, Volkoff, Tolman, and Chandrasekhar, we know that if a star has enough mass, its final and unavoidable fate is to become a black hole [7].

[8] Here I am setting Penrose's singularity theorem as the endpoint of the theoretical debate, in 1965. In [7], Thorne comments that the definitive argument that converted Wheeler was Finkelstein's coordinates in 1958, reproducing a result by Eddington but interpreting it correctly.

There is no force in nature capable of stopping *gravitational collapse*. Presently, black holes are not a theoretical prediction; their existence has been proven experimentally repeatedly. One of the most recent and impressive scientific pieces of evidence is the image of a black hole, in Fig. 4.

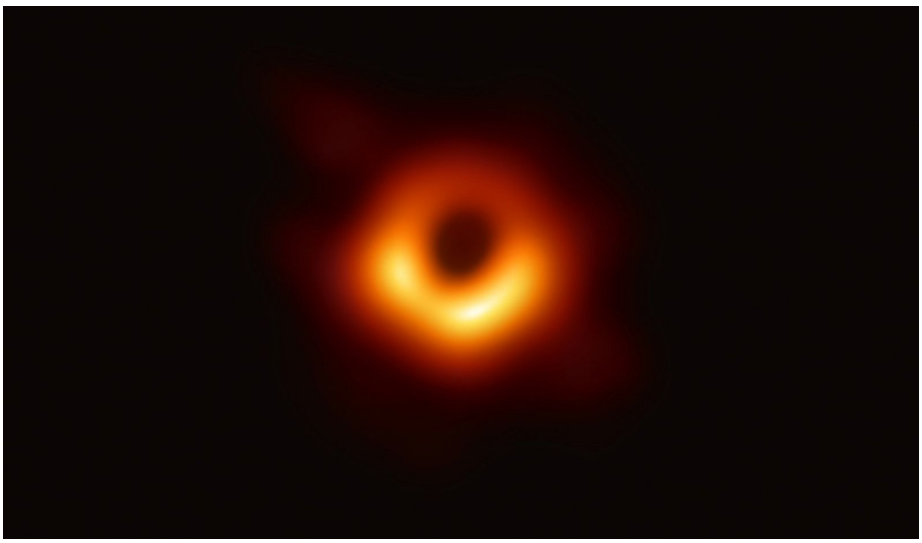

**Fig. 4**: First image of the supermassive black hole in the center of the M87 galaxy, obtained as a result of the *Event Horizon Telescope* collaboration [8].

What we see in Fig. 4 is the supermassive black hole in Messier 87, a galaxy about 53 million light-years from Earth. It was on the cover of newspapers and magazines worldwide. Such black hole is estimated to have a mass of 6.5 billion times the mass of the Sun, a *supermassive* black hole. They receive the surname *supermassive* because they have masses of the order of millions of solar masses.

Unlike astronomical mass black holes, we do not know exactly how they form, but there is substantial evidence that they are found at the centers of galaxies. In colors ranging from orange to yellow we see the black hole's accretion disk, that cluster of matter in the form of hot gas, dust, and plasma, spinning around it.[9]

The black zone we see inside the accretion disk is called the black hole's *shadow*. For a spinning black hole, it is approximately twice the size of the event horizon, and we don't see it directly. This occurs because close to the black hole, light follows curved paths, so much more light than we might initially imagine enters the event horizon and never comes back out, leaving behind a much larger dark zone. Both the size of the ring and how matter behaves around it are in perfect agreement with what General Relativity predicts and represent one of the best current verifications of General Relativity. About five years later, we obtained a similar image of our own planet's supermassive black hole: Sagittarius A*.

[9]The only way matter, primarily in the form of gas and dust, can orbit around and so close to a black hole is by spinning extremely fast, at speeds close to the speed of light. The collision and friction of the different particles cause them to heat up to millions of degrees and emit light of all frequencies.

### 3.3 Tidal forces and spaghettification

Far from a black hole, gravity behaves just like it does around any other body. However, near a black hole's event horizon, fascinating effects start to occur: time begins to pass more slowly on our clocks; if the black hole is rotating, we are forced to rotate with it; and gravity deforms and tears apart massive bodies like stars and other planets. All of this happens because the intensity of gravity increases drastically as we get closer to the event horizon.

One of the most important effects of changing gravity, which also plays a huge role in our daily lives, is *tidal forces*. Remember that for both Newton and Einstein, the magnitude of gravity varies with distance to the attracting body. When we consider a very large body, or an extreme change in gravity over a very small space, there can be considerable differences in acceleration between two points on the same body. The difference in the gravitational pull of the Moon on different points of our planet is the main contributor to tides, hence the name *tidal forces*.

If the tidal force acting on a body is greater than the forces holding it together, it deforms or even tears apart. To build an intuition about tidal forces, we can use Newtonian mechanics and approximations. Consider two points at a distance $d$ from each other, at a distance $r$ from an attracting body of mass $M$. We will assume that $r \gg d$. Using Newton's second law and the potential $V(r) = -mk/r$, the acceleration at a distance $r$ is given by

$$a(r) = \frac{F(r)}{m} = -\frac{1}{m}\frac{dV}{dr} = \frac{k}{r^2}, \tag{3.8}$$

where $k = GM$. If one part of the body is at a distance $r$ and the other part at a distance $r + d$, we find that the difference between the acceleration experienced by each of the points is given by

$$\Delta a = a(r) - a(r+d) = k\left[\frac{1}{r^2} - \frac{1}{(r+d)^2}\right] = k\left[\frac{1}{r^2} - \frac{1}{r^2}\left(1+\frac{d}{r}\right)^{-2}\right]. \tag{3.9}$$

For $r \gg d$, we can use the binomial approximation[10] $(1+p)^{-n} \sim 1 - np + \mathcal{O}(p^2)$ to rewrite

$$\left(1+\frac{d}{r}\right)^{-2} \sim 1 - \frac{2d}{r} + \mathcal{O}\left(\frac{d^2}{r^2}\right), \tag{3.10}$$

and substituting into (3.9), we have[11]

$$\Delta a = k\left[\frac{1}{r^2} - \frac{1}{r^2}\left(1+\frac{d}{r}\right)^{-2}\right] \simeq k\left[\frac{1}{r^2} - \frac{1}{r^2}\left(1-\frac{2d}{r}\right)\right] = \frac{2k\,d}{r^3}. \tag{3.11}$$

The funny thing is that if we employ the Schwarzschild solution and are solely concerned with *radial* acceleration, the result from General Relativity is the same. In both cases, the difference in acceleration between the two points depends on the distance cubed $r^3$, while

[10] This is the Taylor series truncated at the first correction term.

[11] Depending on how you pose the problem, sometimes the factor of 2 isn't present. The important thing here is that the result is proportional to $1/r^3$.

the acceleration of a point mass (or the center of mass of a body) depends on $r^2$. That's why we can see these effects.

In worksheet 3, we do some simple calculations with tidal forces. In our day-to-day life on Earth, we don't even feel them. But the situation gets complicated when I can get too close to a black hole. Due to tidal forces, black holes rip stars apart and turn us into spaghetti if we get too close. If we were swallowed by a black hole of a few solar masses, as we approach its horizon we would be stretched from head to toe, turning into spaghetti.

### 3.4 Orbits in the Schwarzschild solution

As a grand finale, we are going to use General Relativity to calculate the orbits of planets around the Schwarzschild solution. Unlike classical mechanics, the starting point in General Relativity is not an equation as (2.1). Instead of considering gravity as a force acting on a body, we consider how gravity curves space. The curvature of space is what determines how objects move, following trajectories that involve the shortest possible path. These trajectories are called *geodesics*.

For the Schwarzschild metric, that is, around a non-rotating body of mass $M$, we have that the trajectory of a body of mass $m > 0$ is fixed by[12] [2]

$$\frac{1}{2}m\dot{r}^2 + \frac{1}{2}\left(1 - \frac{2GM}{rc^2}\right)\left(mc^2 + \frac{l^2}{mr^2}\right) = \frac{\lambda^2}{2mc^2}, \tag{3.12}$$

where $\lambda$ is a constant: the total relativistic energy.[13] Rearranging this equation, we get

$$\frac{1}{2}m\dot{r}^2 + \frac{mc^2}{2} - \frac{mk}{r} + \frac{ml^2}{2r^2} + \frac{mkl^2}{c^2r^3} = \frac{\lambda^2}{2mc^2}, \tag{3.13}$$

Wait a minute, does this ring any bells? If I define $E = \lambda^2/2mc^2$ (the product of constants is another constant) we have

$$\frac{1}{2}m\dot{r}^2 + \frac{mc^2}{2} - \frac{mk}{r} + \frac{ml^2}{2r^2} + \frac{mkl^2}{c^2r^3} = E. \tag{3.14}$$

This is the equation (2.17) from Newtonian mechanics with several additional terms. That is, by analogy, instead of the effective potential (2.22), we have the effective potential

$$V_{\text{eff,GR}}(r) = \frac{mc^2}{2} - \frac{mk}{r} + \frac{ml^2}{2r^2} - \frac{mkl^2}{c^2r^3}. \tag{3.15}$$

This results in an additional constant term and another proportional to $1/r^3$ in our potential. This seemingly harmless term has many implications. Look at the graph Fig. 5. As we approach $r \to 0$, the behavior of the orbit changes completely. Aside from the stable point of the circular orbit, we now have an unstable point, a maximum, that we can calculate.

[12] Here I am omitting quite a few details for the sake of clarity. In General Relativity, there is no single time: depending on the frame of reference, each observer's measurement of time can vary. However, for bodies with mass $m > 0$, there is a preferred time: *proper time*, which is the $t$ I write in this equation. This is the time that would be measured by a clock traveling attached to the body of mass $m > 0$, or by ourselves with the watch on our wrists if we are the ones orbiting. For weak gravitational fields and velocities far below lightspeed, the difference between observers' times vanishes and we recover our intuition of a single absolute time.

[13] For our purposes, this is a constant. Unlike classical mechanics energy, this energy also includes rest energy: $E_0 = mc^2$, from which Einstein's famous equation comes.

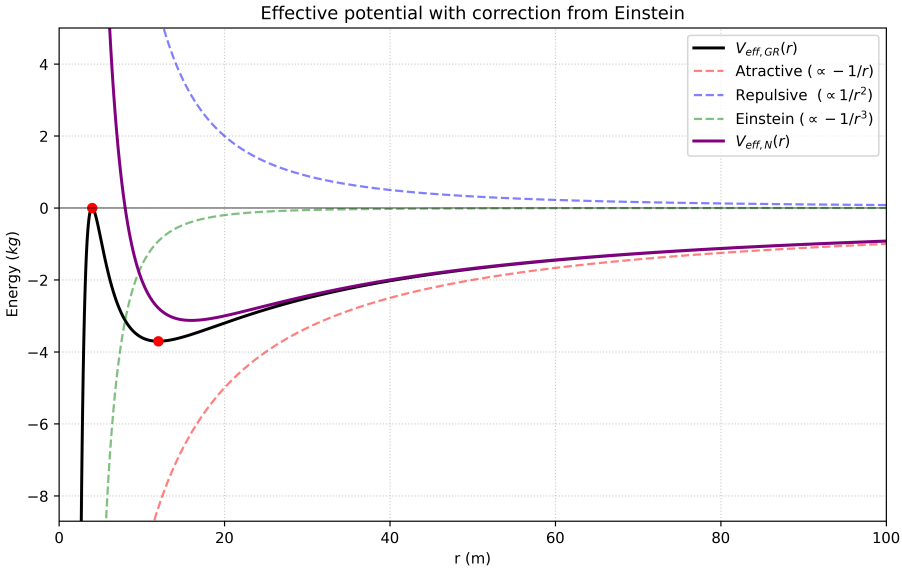


**Fig. 5**: Representation of the potential (3.15), for $c = k = 1$, $m = 100$, and $l = 4k/c$ (solid black line) and comparison with the Newton potential (2.22) (solid violet line). Also depicted are the attractive term (dashed red line), the repulsive term (dashed blue line), and the term proportional to $1/r^3$ from General Relativity. The minimum and maximum of the function $V_{\text{eff}}(r)$ are indicated by red circles. The energy is measured in $kg$ because we have set $c = 1$.

**Exercise 3.2.** *At what radius does the potential become 0? For what values of $l$ do bounded orbits exist? At what radius is the maximum of the potential? What happens for $r$ less than that radius?*

This maximum indicates a point of no return: once the smaller body gets close enough to the larger body, it is doomed to plunge towards $r \to 0$. While in the Newtonian potential (2.22), the term that behaves as $1/r^2$ is positive and dominates the potential for $r \to 0$, by including a negative term proportional to $1/r^3$, we completely change what happens when we bring two bodies together.

But this is not the only change this term introduces. We are going to try to obtain $r(\theta)$ again, but this time using $V_{\text{eff,GR}}(r)$. We have that

$$\frac{d^2u}{d\theta^2} + u - \frac{3k}{c^2}u^2 = \frac{k}{l^2}. \tag{3.16}$$

There is no analytical expression that serves as a solution to this equation, but we can approximate it and see the effects of the approximation. Defining the dimensionless parameter

$$\beta = \frac{3k^2}{l^2c^2}, \tag{3.17}$$

we get the equation

$$\frac{d^2u}{d\theta^2} + u - \frac{\beta l^2}{k}u^2 = \frac{k}{l^2}. \tag{3.18}$$

We will assume that $\beta$ is small (i.e., the Newtonian limit $c^2 \to \infty$) and that we can find a solution to this equation in the form

$$u = u_0 + \beta u_1 + \mathcal{O}(\beta^2), \tag{3.19}$$

where we are only looking for the first term multiplied by $\beta$, assuming the rest are negligible (again, like with the harmonic oscillator). We have that

$$\beta^0 : \frac{d^2 u_0}{d\theta^2} + u_0 = \frac{k}{l^2}, \tag{3.20}$$

$$\beta^1 : \frac{d^2 u_1}{d\theta^2} + u_1 = \frac{l^2}{k} u_0. \tag{3.21}$$

But we already know $u_0(\theta)$ from our Newtonian calculation. Rewriting (2.30) slightly for convenience, we arrive at

$$u_0(\theta) = \frac{k}{l^2}(1 + e\cos\theta), \tag{3.22}$$

so (3.21) becomes

$$\frac{d^2 u_1}{d\theta^2} + u_1 = \frac{k}{l^2}\left[\left(1 + \frac{e^2}{2} + 2e\cos\theta + \frac{e^2}{2}\cos 2\theta\right)\right]. \tag{3.23}$$

And what is the solution to this?

**Exercise 3.3.** *Show, by substitution, that the solution to* (3.23) *is*

$$u_1(\theta) = \frac{k}{l^2}\left[\left(1 + \frac{e^2}{2}\right) + e\theta\sin\theta - \frac{e^2}{6}\cos 2\theta\right]. \tag{3.24}$$

We could keep deriving corrections in $\beta$, but this one is more than enough to see a new effect arising from General Relativity. Here there is a very interesting term: $\theta\sin\theta$. This term **is not periodic**. Once again, we can directly draw this using Python and see what happens, as I propose in worksheet 2. But we can also opt for the analytical route. We will take a fixed point in the orbit and see what happens with this point. Assuming we have an orbit around the Sun, let's focus on the point where the orbiting body is closest to the Sun. This point is named the *perihelion* (from Greek *peri-*, meaning *near*, and *-helion*, meaning Sun). When we are at the perihelion, $r$ is at a minimum, which implies that $u = 1/r$ is at a maximum. Therefore, the perihelion is given by

$$\frac{du}{d\theta} \simeq \frac{d(u_0 + \beta u_1)}{d\theta} = 0 \implies -e\sin\theta + \beta\left(e\sin\theta + e\,\theta\cos\theta + \frac{e^2}{3}\sin(2\theta)\right) = 0. \tag{3.25}$$

Inspecting the terms, we have a solution for $\theta = 0$. Since sine and cosine have a period of $2\pi$, we know that the next position of the perihelion will be for $\theta = 2\pi + \delta$, where $\delta$ is a small amount. To estimate $\delta$, we first substitute $\theta = 2\pi + \delta$ into the equation and use the small-angle approximations: $\cos(2\pi + \beta) \sim 1 + 2\pi + \beta + \mathcal{O}(\beta^2)$ and $\sin(2\pi + \beta) \sim 2\pi + \beta + \mathcal{O}(\beta^3)$. Discarding terms of order $\delta^2$ and $\beta\delta$, we arrive at

$$\delta = 2\pi\beta = \frac{6\pi k^2}{l^2 c^2}. \tag{3.26}$$

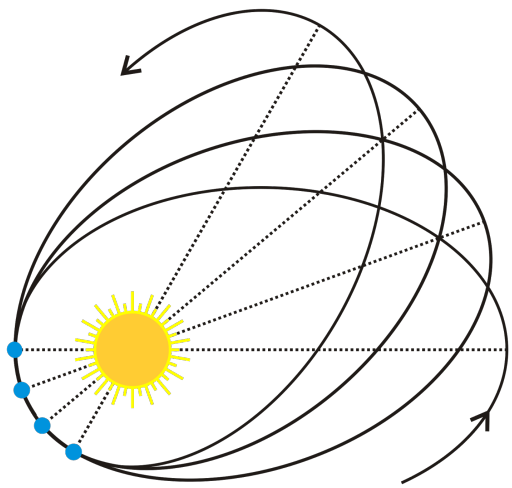

**Fig. 6**: Illustration of the precession of Mercury's orbit, from Carlos Roque Sánchez. Mercury's orbits around the Sun are not periodic, but rotate around the Sun, keeping our star at the focus of the ellipse.

That is to say, planets do not follow closed orbits. Instead, the orbit itself rotates, as illustrated in Fig. 6. This rotation of the orbit is known as precession. As (3.26) indicates, this only depends on the angular momentum $l$ and the mass of the attracting body, in our case the Sun.

For Mercury, the planet closest to the Sun, we can estimate the precession of its orbit. We know that Mercury has $e \sim 0.2$ and that the aphelion is at $r_+ \sim 7 \times 10^{10}\ km$. Using our Newtonian result (2.31), we have that the aphelion is given by

$$r_+ = \frac{l^2}{k}\frac{1}{1-e} \implies l^2 = k\, r_+(1-e). \tag{3.27}$$

Substituting into (3.26) we have that

$$\delta = \frac{6\pi k}{c^2}\frac{1}{r_+(1-e)}. \tag{3.28}$$

Using the values for Mercury, we obtain a precession of $\delta \sim 5.0 \times 10^{-7}$ radians per orbit. This seems tiny, but it accumulates over the years. In one century, Mercury circles the Sun 415 times, giving a precession of $2.1 \times 10^{-4}$ radians per century. In astronomical units, this is about $43''$ (43 *arcseconds*). This is one of the most important predictions of the theory of General Relativity.[14]

At the end of worksheet 2, I propose an exercise similar to the one we just did, but this time, for the orbits followed by particles of light: *photons*. In General Relativity, photons follow another type of trajectory, *null geodesics*, which can be calculated with another effective potential.

[14]The actual precession of Mercury's perihelion is much greater, approximately $575''$. Our approximation is much smaller because we are ignoring the effect of other planets. That calculation is much more complex and is written in detail in [2].

We have completed a journey that goes from Newton's intuition to the complexity of Schwarzschild. Your role now is to encourage students not to see these formulas as static entities, but as living tools that, thanks to programming in Python, can come to life on their screens. I hope this manual serves as an inspiration to transform the classroom. Students' curiosity about black holes is the perfect gateway to consolidate the more rigorous physics and mathematics concepts in High School.


## Acknowledgments

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## Bibliography


[1] D. Tong, *Part IA, Dynamics and Relativity*, *University of Cambridge* (2013) .

[2] D. Tong, *Part III, General Relativity*, *University of Cambridge* (2019) .

[3] H. Reall, *Part III, General Relativity*, *University of Cambridge* (2013) .

[4] R.M. Wald, *General Relativity*, Chicago Univ. Pr., Chicago, USA (1984), 10.7208/chicago/9780226870373.001.0001.

[5] J.H. Rogers, *Origins of the ancient constellations: I. The Mesopotamian traditions*, *Journal of the British Astronomical Association* **108** (1998) 9.

[6] C. Sagan, *Cosmos*, Planeta, Barcelona, España (1980).

[7] K.S. Thorne, *Black holes and time warps: Einstein's outrageous legacy*, W. W. Norton & Company (1994).

[8] Event Horizon Telescope collaboration, *First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole*, *Astrophys. J. Lett.* **875** (2019) L1 [1906.11238].